# The first law

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a consistent sign convention in an energy balance.

**SUCCESS CRITERIA - I can...**

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “internal energy” and “work” to explain your model or result. Sentence starters are optional.

**internal energy**: energy stored in a system’s microscopic motion/interactions

**work**: energy transferred by a force along a displacement

**surroundings**: everything outside the chosen system

## 01 Bring back an earlier idea

A 3 kg sample completely changes phase with latent heat 500 J/kg.
**Find** the energy magnitude required.

**Word help:** latent heat = energy per mass needed for a specified phase change

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $\Delta U = Q - W_{by}$.
Q is positive into the system; $W_{by}$ is positive when the system does work on its surroundings.

**Worked reasoning:** A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

**Your turn: Draw** labeled arrows crossing a system boundary and mark the 60 J stored increase.

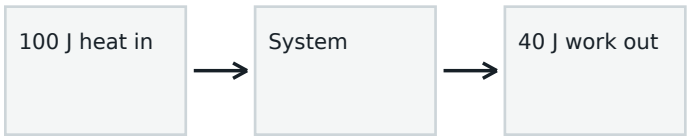
**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

A system absorbs 100 J of heat and does 40 J of work.
**Draw** the energy arrows and find its internal-energy change.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** ① ② ③

## 04 Build the solution

A system receives 80 J and does 20 J of work. Complete ΔU=80-20=____ J. Does internal energy increase or decrease?

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

Why does the phrase “work is 60 J” need a direction or sign convention?

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

Total kinetic energy changes from 24 J to 20 J in an isolated collision. Is it elastic? Where could the difference go?

**Word help:** kinetic energy = energy associated with motion

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

Heat input 30 J; work output 10 J.
**Find** ΔU.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

**Give** two different heat-input/work-output pairs that increase internal energy by 40 J.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3